MTH-202

Quiz file 4

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BY ATTIQ KUNDI

1. Which of the following is correct option?

Answer: C (n, 0) =1

2. Which of the following is correct option?

Answer: C (n, 1) =n

3. Number of distinct permutations that can be formed using the letters of the word **"Bubble"** are____.

Answer: 120

4. If A and B are disjoint sets then the mathematical representation of inclusion-exclusion principle.

Answer: n (AUB) = n (A) + n (B)

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5. If A is only set and U is the universal then

N (A') = ____.

Answer: n (U) – n (U∩A)

6. There are 5 girls' students and 20 boys students in a class. How many students are there in total?

Answer: 25

7. If A and B are disjoint finite sets then

n (AUB) = ____.

Answer: n (A) + n (B)

8. If A and B are finite set then

n (AUB) = ____.

Answer: n (A) +n (B) – n($A \cap B$)

9. Find the number of the word that can be formed of the letters of the word **"ELEVEN"**

Answer: 120

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10. C (n, k) = k! P (n, k).

Answer: false

11. In which century the theory of probability was first developed?

Answer: 17th centaury

12. C (n, 0) = ___

Answer: 1

13. A boy have 5 res and 7 blue balls, what is probability that a ball chosen from the boy is red?

Answer: 5/12

14. **C (n, 1) = ____.**

Answer: n

15. Let X = {1, 2, 3}, then 2-combinations of the 3 elements of the set X are ____.

Answer: {1, 2}, {1, 3} and {2, 3}

16. Let a and B be subsets of U with n (A) = 12, n (B) = 15, n (A') = 17, and n (A intersection B) = 8, then n (U) = ____.

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Answer: 29

17. In a letter, players win the prize when they pick two digits that match, in the correct order, two digits kept secret. Which of the following is the probability of wining the prize?

Answer: 1/100

18. A tree is normally constructed from ____.

Answer: left to right

19. Find the value of C(n, n)

Answer: n

20. How many different signals each consisting of five flags hung in a vertical line, can be formed from three identical red flags and two identical blue flags?

Answer: 10

21. Let n (A) = 108, n (B) = 72, and n $(A \cup B) =$ ___

Answer: 144

22. If A and B are finite sets then the mathematical representation of inclusion-exclusion principle.

Answer: n (AUB) = n (A) + n (B) –n$(A \cap B)$.

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23. If k=10 and n=2; find the number of ways using K-selection.

Answer: 11.

24. What is the minimum number of students in a class to be sure that two of them are born in the same month?

Answer: 13.

25. Which of the following is the product set A*B*C? Where A = {a}, B = {b}, and C = {c, d}.

Answer: {(a, b, c), (a, b, d)}

26. 0! = _____.

Answer: 1

27. Find the number of distinct permutations that can be formed using the letters of the word **"BENZENE".**

Answer: 420

28. If two light bulbs are chosen at random from 5 bulbs of which 3 are defective, then which of the following is the probability that none is defective?

Answer: 2/8

29. When a dice and a coin are tossed together, then which of the following is the sample space?

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Answer: {1H, 2H, 3H, 4H, 5H, 6H, 1T, 2T, 3T, 4T, 5T, 6T}

30. Which of the followings is the number of distinct permutations that can be formed using the letters of the word BUZZ?

Answer: 12

31. The probability of a an event lies between 0 and 1.

Answer: true

32. How many integers from 1 through 1002 are multiples of 4?

Answer: 250.5

33. The number of the words that can be formed from the letters of the word, "COMMITTEE" are.

Answer: 9! / (2! 2! 2!)

34. The number of possible permutations of the letters of the word, "ADDING" having two D's together.

Answer: 5!

35. _____ is a procedure that yields a given set of possible outcomes?

Answer: experiment

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36. If a die is tossed how many outcomes the sample space of the experiment will have?

Answer: 6

37. In how many ways can the letters of the word "LEADER" be arranged?

Answer: 360

38. An urn contains six red and nine blue balls. What is the probability that a ball chosen from the urn is blue?

Answer: 9/15

39. Among 20 people, 15 either swim or jog or both. If 5 swim and 6 swim and jog. How many jog?

Answer: 16.

40. A boy have 3 red and 4 blue balls, what is the probability that a ball chosen from the boy is blue?

Answer: 3/7

41. If a pair of dice is tossed how many outcomes the sample space of the experiment will have?

Answer: 36

42. Who gave the first definition of the probability as the number of successful outcomes divided by the number of total outcomes?

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Answer: Laplace

43. In a lottery, players win the prize when they pick two digits that match, in the correct order, two digits kept secret. Which of the following is the probability of wining the prize?

Answer:

44. Let X = {1, 2, 3}, then 2-combinations of the 3 elements of the set x are ____.

Answer: {1, 2}, {1, 3} and {2, 3}

45. An event is a subset of?

Answer: sample space

46. The ____ of the experiment is the set of possible outcomes.

Answer: sample space

47. A student is to answer five out of nine questions on exams. Find the number of ways that can choose the five questions.

Answer: 126.

48. Suppose that A and B are events in a sample space. If A and b disjoint, could P (A) = 0.6 and P (B) = 0.5?

Answer: No

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49. If 8 people are divide into teams of size 2, 3, 4. Find the number of possible arrangements.

Answer: 240

50. In how many ways can a set of six letters be selected from the ENGLISH Alphabets?

Answer: C (26, 6)

51. What is the probability of getting a number greater than 5 when a dice is tossed?

Answer: 1/6

52. How many bit strings of length two do not have two consecutive 1's?

Answer: 2

53. Suppose a box B contain four numbered 1 through 4. Which of the followings is the number m of ways of from B firstly three marbles, then remaining one marble?

Answer: 27

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